PROJECT REPORT

ON

# FINSIGHT — FINANCE ANALYTICS DASHBOARD

*Real Time Insight into Transaction, Customers & Risk*

| 45.32M | 4.99K | 9.08K | 75.44K | 13.57K |
|---|---|---|---|---|
| **Total Amount** | **Total Transactions** | **Avg Txn Value** | **Total Fee** | **Total Tax** |
| *All transactions* | *Unique transactions* | *Per transaction* | *Transaction fees* | *Tax collected* |

**Submitted By:**

Adarsh Shandilya

# ABSTRACT

This project presents FinSight — a comprehensive Finance Analytics Dashboard built on two real financial datasets: a Customers dataset (5,000 records, 11 attributes) and a Finance Transactions dataset (50,069 records, 15 attributes). The solution delivers real-time insight into transaction performance, customer behaviour, risk metrics, fees, and tax collection through an interactive two-page Power BI report.

The analytical pipeline covers data cleaning (fixing name casing, handling duplicate channel labels like M@bile App and Mobile App, standardising date formats), feature engineering (deriving Total Amount, Total Tax, Total Fee, YoY growth metrics), and DAX-powered calculations for five core KPIs: Total Amount (Rs.45.32M), Total Transactions (4.99K), Average Transaction Value (Rs.9.08K), Total Fee (Rs.75.44K), and Total Tax (Rs.13.57K).

The two-page dashboard includes an Overview Analysis page featuring monthly tax trend, transaction status donut, customer segment bar, state-wise bar, transaction type matrix, and gender donut; and a Transactions page featuring a detailed tabular drill-through of all individual transactions with filters for Year, Dynamic Metrics, Occupation, and Category. Key findings include Loan EMI as the highest-volume transaction type (9,140 transactions, Rs.12.31M amount), Maharashtra leading state revenue (Tax Rs.2.1K), Success rate at 85.2% of all transactions, Retail segment dominating at Rs.7K tax, and Male customers slightly outnumbering Female (53.3% vs 46.7%).

# TABLE OF CONTENTS



*Table 1: Table of Contents*

# CHAPTER 1: INTRODUCTION

## 1.1 Financial Analytics and Business Intelligence

The global financial services industry generates billions of transactions daily. For banks, NBFCs, and fintech companies, real-time visibility into transaction patterns, customer behaviour, fee collection, and risk metrics is essential for regulatory compliance, fraud prevention, and business growth. According to Deloitte (2023), financial institutions that adopt advanced analytics reduce operational losses by 20-30% and improve customer retention by 15-25%.

Power BI has emerged as the leading Business Intelligence tool for financial dashboards, offering DAX-powered calculations, interactive slicers, drill-through reports, and real-time data refresh capabilities. This project — FinSight — demonstrates the complete pipeline from raw financial CSV data to a production-grade interactive analytics dashboard.

## 1.2 Project Overview

FinSight is a two-page Power BI dashboard built on real transaction and customer data from a financial organisation. The system processes 50,069 transaction records linked to 5,000 customers across multiple states, segments, and transaction types. The dashboard enables business analysts and management to monitor KPIs, track trends, and drill into individual transactions — all with interactive filtering by Year, Occupation, Category, and Dynamic Metrics.

## 1.3 Dataset Description

| Dataset | Records | Columns | Key Information |
|---|---|---|---|
| Customers | 5,000 | 11 | customer_id, name, gender, DOB, city, state, occupation, customer_segment, annual_income, join_date |
| Finance Transactions | 50,069 | 15 | transaction_id, date, customer_id, transaction_type, channel, amount, fee, tax, status, is_fraud, risk_score |

*Table 2: Dataset Summary*

## 1.4 Business Context

| Dimension | Details |
|---|---|
| Customer Base | 5,000 customers across 5 segments: Retail (2,703), Premium (895), SME (780), Corporate (374), Wealth (248) |
| Transaction Volume | 50,069 transactions in Rs.456M total (raw); Dashboard shows filtered 2026 view: Rs.45.32M |
| Transaction Types | 10 types: Loan EMI (9,140), Transfer (8,479), Card Payment (5,329), Deposit (4,970), Bill Payment (4,310), others |
| Transaction Status | Success: 42,930 (85.7%) \| Failed: 5,103 (10.2%) \| Pending: 2,036 (4.1%) |
| Geography | 5+ states: Maharashtra, Gujarat, UP, Karnataka, Tamil Nadu, MP, Haryana, Kerala, Rajasthan, Delhi |
| Channels | UPI, Branch, ATM, Net Banking, POS, Auto Debit, Mobile App (7 clean channels) |
| Gender Split | Female: 2,544 (50.9%) \| Male: 2,456 (49.1%) — customers; Dashboard shows Male 53.3% of tax |

*Table 3: Business Context and Dataset Dimensions*

# CHAPTER 2: OBJECTIVES OF THE PROJECT

## 2.1 Primary Objectives

1. To build an end-to-end Finance Analytics Dashboard (FinSight) on 55,069 financial records (5,000 customers + 50,069 transactions) delivering real-time KPIs and interactive visualisations.
2. To track 5 core KPIs for 2026: Total Amount (Rs.45.32M), Total Transactions (4.99K), Avg Transaction Value (Rs.9.08K), Total Fee (Rs.75.44K), Total Tax (Rs.13.57K) with YoY comparison.
3. To design two specialised dashboard pages: Overview Analysis for aggregate insights and Transactions page for individual drill-through analysis.
4. To implement dynamic filtering via Year, Dynamic Metrics, Occupation, and Category slicers enabling personalised financial analysis for different user roles.

## 2.2 Secondary Objectives

5. To analyse 10 transaction types (Loan EMI, Transfer, Card Payment, Deposit, Bill Payment, Withdrawal, Interest Credit, Fee Charge, Investment, Refund) by amount, tax, and fee to identify revenue and cost drivers.

6. To assess transaction success rates (85.2% Success, 10.9% Failed, 3.9% Pending) and identify failure patterns by type and segment.
7. To compare 5 customer segments (Retail, Premium, SME, Corporate, Wealth) by tax contribution, transaction volume, and average transaction value.
8. To perform state-wise analysis identifying Maharashtra, Gujarat, and Uttar Pradesh as top revenue states.
9. To analyse gender-based transaction patterns and identify demographic opportunities for targeted financial products.

# CHAPTER 3: TOOLS AND REQUIREMENT SPECIFICATIONS

## 3.1 Software Requirements

| Software | Version | Purpose |
|---|---|---|
| Microsoft Power BI Desktop | 2024 Latest | Primary dashboard tool: data model, DAX, slicers, visuals |
| Microsoft Excel | 2019/365 | Source data files (customers.csv, finance_transactions.csv) |
| Power Query (built-in) | Latest | Data cleaning: name casing, date conversion, channel deduplication |
| DAX (built-in) | Latest | KPI measures: YoY, Total Amount, Tax, Fee, Avg Transaction Value |
| Python (Optional) | 3.10+ | Pre-processing, EDA, additional chart generation |
| Pandas | 2.0+ | Data exploration and cleaning validation |
| Jupyter Notebook | 6.5+ | Python EDA environment |
| Git / GitHub | Latest | Version control and portfolio hosting |

*Table 4: Software Requirements*

## 3.2 Hardware Requirements

| Component | Minimum | Recommended |
|---|---|---|
| Processor | Intel i3 8th Gen | Intel i5/i7 10th Gen+ |
| RAM | 8 GB DDR4 | 16 GB DDR4 |
| Storage | 256 GB SSD | 512 GB SSD |
| Display | 1366x768 | 1920x1080 FHD (for dashboard design) |
| OS | Windows 10 64-bit | Windows 11 64-bit |

*Table 5: Hardware Requirements*

# CHAPTER 4: PROBLEM DEFINITION

## 4.1 Business Problems Identified

| # | Problem | Evidence from Data | Business Impact |
|---|---|---|---|
| P1 | No YoY performance visibility | Total Amount - Rs.539.14K vs previous year (Dashboard KPI) | Cannot identify declining revenue trends early |
| P2 | Transaction failure rate (10.2%) | 5,103 Failed of 50,069 total transactions | Revenue leakage; customer dissatisfaction; re-processing cost |
| P3 | Channel data quality issues | Duplicate channel labels: M@bile App, Mobile App, extra spaces in Net Banking | Incorrect channel analytics; misleading segmentation |
| P4 | No segment-level profitability view | 5 customer segments with no comparative dashboard | Cannot identify which segments drive fees/tax vs cost |
| P5 | Pending transactions (4.1%) | 2,036 Pending transactions unresolved | Liquidity risk; unrecognised revenue; compliance issue |
| P6 | State revenue concentration | Maharashtra, Gujarat, UP = top 3 states (Rs.2.1K + Rs.1.7K + Rs.1.7K tax) | Geographic concentration risk; under-served markets |

*Table 6: Identified Business Problems with Data Evidence*

## 4.2 Technical Problem Statement

The raw datasets contain multiple data quality issues: customer names in inconsistent mixed case (e.g., BobbY JacksoN), transaction channels with duplicate labels (M@bile App vs Mobile App, leading/trailing spaces), date fields stored as strings, missing feature columns (Total Amount = amount + fee + tax not pre-calculated), and no YoY baseline for comparison. A systematic Power Query transformation pipeline is required before dashboard modelling.

## 4.3 Business Requirements Summary

As specified in the Business Requirements document provided, the financial organisation requires a centralised analytics solution to: monitor KPIs in real time, identify high-performing customer segments and states, analyse transaction patterns and trends, track operational fees and taxes, understand customer demographics, and improve financial decision-making. The dashboard must allow dynamic filtering by Year, Dynamic Measure, Occupation, and Category.

# CHAPTER 5: REQUIREMENT SPECIFICATIONS

## 5.1 Functional Requirements

| ID | Requirement | Priority | Status |
|---|---|---|---|
| FR-01 | Import customers.csv (5,000 records) and finance_transactions.csv (50,069 records) into Power BI | High | Done |
| FR-02 | Clean data: fix name casing, standardise channel names, convert date strings | High | Done |
| FR-03 | Create calculated column: Total = amount + fee_amount + tax_amount | High | Done |
| FR-04 | Build DAX measures: Total Amount, Total Transactions, Avg Transaction Value, Total Fee, Total Tax | High | Done |
| FR-05 | Build YoY comparison DAX for all 5 KPIs | High | Done |
| FR-06 | Create KPI cards with trend indicators on Page 1 (Overview) | High | Done |
| FR-07 | Build area chart: Total Tax by Month (Jan-Apr 2026 trend visible) | High | Done |
| FR-08 | Build donut chart: Total Tax by Transaction Status (Success 85.2%, Failed 10.9%, Pending 3.9%) | High | Done |
| FR-09 | Build horizontal bar: Total Tax by Customer Segment (Retail, Premium, SME, Corporate, Wealth) | High | Done |
| FR-10 | Build horizontal bar: Total Tax by State (10 states shown) | High | Done |
| FR-11 | Build matrix table: Transaction Type Analysis (Type x Tax, Fee, Amount) | High | Done |
| FR-12 | Build donut chart: Total Tax by Gender (Female 46.7%, Male 53.3%) | High | Done |
| FR-13 | Build Page 2 (Transactions): tabular drill-through of all transactions with 9 columns | Medium | Done |
| FR-14 | Add slicers: Year, Dynamic Metrics, Occupation, Category on left navigation panel | High | Done |
| FR-15 | Implement page navigation: Overview Analysis button and Transactions button | Medium | Done |

*Table 7: Functional Requirements with Completion Status*

## 5.2 Non-Functional Requirements

| ID | Requirement | Target |
|---|---|---|
| NFR-01 | Performance | Dashboard loads in < 5 seconds; slicer filtering in < 2 seconds |
| NFR-02 | Accuracy | All KPI values verified against source data calculations |
| NFR-03 | Usability | Finance team can navigate and filter without technical training |
| NFR-04 | Scalability | Power BI model handles up to 500K transaction records without redesign |
| NFR-05 | Maintainability | New monthly data refreshable by updating source CSV files only |

*Table 8: Non-Functional Requirements*

# CHAPTER 6: PROJECT PLANNING AND SCHEDULING

## 6.1 Gantt Chart

| Phase | Activity | Duration | Weeks | Deliverable |
|---|---|---|---|---|
| Ph.1 | Requirement Analysis & Dataset Study | 7 days | 1 | Business Requirements Doc + Data Dictionary |
| Ph.2 | Data Cleaning & Power Query Transformation | 5 days | 2 | Cleaned data model in Power BI |
| Ph.3 | DAX Measure Development | 4 days | 2 | KPI measures + YoY calculations |
| Ph.4 | Page 1 — Overview Dashboard Design | 7 days | 3 | 6 visuals + 5 KPI cards + slicers |
| Ph.5 | Page 2 — Transactions Drill-Through | 4 days | 3-4 | Tabular transaction report |
| Ph.6 | Testing & Validation | 3 days | 4 | Values verified vs source data |
| Ph.7 | Documentation & Report Writing | 7 days | 5 | This project report (DOCX) |
| Ph.8 | GitHub Upload & LinkedIn Post | 2 days | 5-6 | Public repository + posts |

*Figure 1: Project Gantt Chart — 6-Week Schedule*

## 6.2 Risk Register

| Risk | Probability | Impact | Mitigation |
|---|---|---|---|
| Duplicate channel labels corrupt analysis | High | High | Clean in Power Query: standardise all channel values to 7 clean types |
| YoY baseline data unavailable | Medium | Medium | Create a reference table or use hardcoded previous year baseline |
| 50K transactions slow Power BI | Low | Medium | Pre-aggregate in Power Query; use Import mode not DirectQuery |
| Dynamic Metrics slicer complexity | Medium | Low | Use Field Parameters feature in Power BI for dynamic measure switching |

*Table 9: Project Risk Register*

# CHAPTER 7: ANALYSIS

## 7.1 Level 0 DFD — Context Diagram

The FinSight system receives inputs from two sources: the Customers CSV (5,000 records) and Finance Transactions CSV (50,069 records), plus User Inputs (slicer selections for Year, Occupation, Category, Dynamic Metrics). It delivers outputs to the Overview Dashboard (aggregate KPIs and charts) and the Transactions Dashboard (individual transaction drill-through).

*Figure 2: Level 0 DFD — Context Diagram (insert from notebook)*

## 7.2 Level 1 DFD

| Process | Name | Input | Output |
|---|---|---|---|
| P1 | Data Ingestion | customers.csv + finance_transactions.csv | Raw tables in Power BI Power Query |
| P2 | Data Cleaning | Raw tables with quality issues | Cleaned tables (proper case, clean channels, correct dates) |
| P3 | Data Modelling | Cleaned tables | Relationship model: Customers (1) to Transactions (M) |
| P4 | Feature Engineering | Modelled tables | Total = amount + fee + tax; YoY calculated columns |
| P5 | DAX Measures | Power BI data model | 5 KPI measures + 5 YoY measures + segment/status measures |
| P6 | Dashboard Design | DAX measures + cleaned data | Page 1 (Overview) + Page 2 (Transactions) .pbix file |

*Figure 3: Level 1 DFD — Process Decomposition*

## 7.3 Level 2 DFD — Data Cleaning (P2)

| Sub-process | Action | Input | Output |
|---|---|---|---|
| P2.1 | Fix Name Casing | Mixed case customer names | Proper case via Text.Proper() |
| P2.2 | Standardise Channel Labels | M@bile App, Mobile App, extra spaces | Clean 7 channel values via Text.Trim + Replace |
| P2.3 | Convert Date Strings | Date as text strings | Date type via Date.From() |

| | | | |
|---|---|---|---|
| P2.4 | Add Total Column | amount, fee_amount, tax_amount columns | Total = amount + fee_amount + tax_amount |
| P2.5 | Extract Year/Month | Transaction date column | Year and Month Name derived columns |
| P2.6 | Handle is_fraud Column | Yes/No text | Binary flag: 1=Fraud, 0=Normal |

*Figure 4: Level 2 DFD — Data Cleaning Sub-processes*

## 7.4 Entity Relationship Diagram

| Entity | Primary Key | Foreign Key | Relationship | Cardinality |
|---|---|---|---|---|
| CUSTOMERS | customer_id | — | One customer has many transactions | 1 : M |
| TRANSACTIONS | transaction_id | customer_id -> CUSTOMERS | Each transaction belongs to one customer | M : 1 |
| CALENDAR (derived) | Date | — | One date links to many transactions | 1 : M |

*Figure 5: Entity Relationship Diagram*

## 7.5 Class Diagram

| Class | Attributes | Key Methods |
|---|---|---|
| DataLoader | file_path, table_name, df | load_csv(), validate_schema(), get_row_count() |
| DataCleaner | raw_df, clean_df | fix_names(), clean_channels(), convert_dates(), add_total_col() |
| FeatureEngineer | df | add_year_month(), add_fraud_flag(), create_calendar_table() |
| DashboardBuilder | pbix_path, data_source | import_data(), create_relationship(), add_measures(), design_pages() |
| ReportGenerator | analysis_results | create_docx(), export_visuals(), build_github_readme() |

*Figure 6: Class Diagram — System Object Model*

# CHAPTER 8: COMPLETE PROJECT STRUCTURE

## 8.1 Module Descriptions and Effort

| Module | File | Description | Effort |
|---|---|---|---|
| M1 | Power Query Steps | Clean name casing, channels, dates, add Total column | 10 hrs |
| M2 | DAX Measures Table | 5 KPI measures + 5 YoY measures + segment/type measures | 15 hrs |
| M3 | Page 1 — Overview | 6 charts + 5 KPI cards + 4 slicers + navigation | 25 hrs |
| M4 | Page 2 — Transactions | Drill-through table + same slicers | 10 hrs |
| M5 | EDA (Python/Jupyter) | Data exploration, channel cleanup verification, stats | 10 hrs |
| M6 | Report & GitHub | DOCX report, README, screenshots, LinkedIn posts | 10 hrs |
| | | TOTAL | 80 hrs |

*Table 10: Module Description and Effort Estimation*

## 8.2 Key DAX Measures

| Measure | DAX Formula | 2026 Result |
|---|---|---|
| Total Amount | SUM(transactions[amount]) | Rs.45.32M |
| Total Transactions | COUNT(transactions[transaction_id]) | 4,990 (4.99K) |
| Avg Transaction Value | DIVIDE([Total Amount],[Total Transactions]) | Rs.9.08K |
| Total Fee | SUM(transactions[fee_amount]) | Rs.75.44K |
| Total Tax | SUM(transactions[tax_amount]) | Rs.13.57K |
| YoY Amount | [Total Amount] - CALCULATE([Total Amount],DATEADD(Calendar[Date],-1,YEAR)) | Rs.-539.14K |
| YoY Transaction Count | [Total Transactions] - CALCULATE([Total Transactions],SAMEPERIODLASTYEAR(Calendar[Date])) | 0.00 |
| YoY Avg Txn Value % | ([Avg Txn Value]/[Prev Year Avg Txn Value]-1)*100 | -1.57% |
| YoY Fee % | ([Total Fee]/[Prev Year Fee]-1)*100 | +7.17% |
| YoY Tax % | ([Total Tax]/[Prev Year Tax]-1)*100 | +7.14% |

| | | |
|---|---|---|
| Success Rate % | DIVIDE(CALCULATE([Total Transactions],transactions[status]='Success'),[Total Transactions])*100 | 85.2% |
| Dynamic Metric | SWITCH(SELECTEDVALUE(MetricParam[Metric]),'Total Tax',[Total Tax],'Total Fee',[Total Fee],...) | Dynamic |

*Table 11: Complete DAX Measures with Formulas and Results*

## 8.3 Process Logic — All Modules

### M1: Power Query Cleaning

Step 1: Load customers.csv — Text.Proper([first_name]) and Text.Proper([second_name]). Step 2: Load finance_transactions.csv — Text.Trim and Text.Replace all channel variants to clean 7 values. Step 3: Date.From([transaction_date]) to convert string dates. Step 4: Add column Total = [amount] + [fee_amount] + [tax_amount]. Step 5: Add Year = Date.Year([transaction_date]). Step 6: Add Month = Date.MonthName([transaction_date]). Step 7: Close and Apply.

### M2: DAX Measures

Step 1: Create blank table 'Measures' to store all measures. Step 2: Create Calendar table using CALENDAR(MIN, MAX of transaction_date). Step 3: Mark Calendar[Date] as Date Table. Step 4: Create relationship Customers[customer_id] -> Transactions[customer_id] (1:M). Step 5: Write all 12 measures as listed in Table 11. Step 6: Create Field Parameter table for Dynamic Metrics slicer. Step 7: Verify each measure value matches source data cross-check.

### M3: Page 1 — Overview Design

Step 1: Set page background to dark navy. Step 2: Add FinSight logo and title bar. Step 3: Add 5 KPI cards row with value and YoY indicator. Step 4: Add area chart: X=Month, Y=Total Tax. Step 5: Add donut chart: Legend=Transaction_Status, Values=Total Tax. Step 6: Add horizontal bar: Y=Customer_Segment, X=Total Tax. Step 7: Add horizontal bar: Y=State, X=Total Tax. Step 8: Add matrix: Rows=Transaction_Type, Values=Transaction Count, Tax, Fee, Amount. Step 9: Add donut: Legend=Gender, Values=Total Tax. Step 10: Add left panel slicers: Year (dropdown), Dynamic Metrics (dropdown), Occupation (dropdown), Category (dropdown). Step 11: Add navigation buttons: Overview Analysis (active) and Transactions.

### M4: Page 2 — Transactions Design

Step 1: Copy slicer panel from Page 1. Step 2: Add navigation buttons (Transactions highlighted). Step 3: Add 5 KPI cards (same DAX, shows filtered values). Step 4: Add table visual with columns: transaction_id, transaction_date, Customer_name, transaction_type, transaction_status, gender, customer_segment, state, Total, Total tax, Total amount. Step 5: Sort by Total amount ascending. Step 6: Verify slicer filtering applies across all table rows.

## 8.4 Deliverables

| # | Deliverable | Format | Contents |
|---|---|---|---|
| 1 | Business Requirements | DOCX | Provided — defines all KPI and chart requirements |
| 2 | Cleaned Dataset | Power BI Internal | Power Query steps applied: casing, channels, dates, Total column |
| 3 | Finance Analytics Dashboard | PBIX | 2-page interactive report with 12 visuals and 4 slicers |
| 4 | Dashboard Screenshots | PNG x2 | Page 1 Overview + Page 2 Transactions |
| 5 | Project Report | DOCX | This complete documentation |
| 6 | GitHub Repository | URL | Public repo: business requirement, datasets, PBIX, screenshots, report |

*Table 12: Complete Project Deliverables*

# CHAPTER 9: DASHBOARD 1 — OVERVIEW ANALYSIS

## 9.1 Dashboard Design

The Overview Analysis page is a dark-themed, single-screen executive dashboard with a navy blue background, gold KPI values, and blue chart elements. The left panel contains four slicers (Year, Dynamic Metrics, Occupation, Category) and two navigation buttons. The main content area displays five KPI cards across the top and six visualisations below. The current view is filtered to Year 2026.

## 9.2 KPI Cards

| 45.32M | 4.99K | 9.08K | 75.44K | 13.57K |
|---|---|---|---|---|
| **Total Amount** | **Total Transactions** | **Avg Txn Value** | **Total Fee** | **Total Tax** |
| *-539.14K vs Prev Year* | *0.00 vs Prev Year* | *-1.57% vs Prev Year* | *+7.17% vs Prev Year* | *+7.14% vs Prev Year* |

*Table 13: Overview KPI Cards — 2026 Filtered Values*

**KPI Analysis and Business Interpretation**

► Total Amount DECLINED by Rs.539.14K vs previous year — revenue contraction alert for management

► Total Transactions = 0.00 change vs previous year — volume flat, but YoY comparison needs validation

► Avg Transaction Value fell 1.57% — customers making smaller transactions in 2026

► Fee revenue INCREASED 7.17% — more transactions attracting processing fees (positive for bank)

► Tax collection INCREASED 7.14% — aligned with fee increase; tax is proportional to fee

► Interpretation: Volume flat + Amount declining + Fee up = customers making more small-value fee-bearing transactions

## 9.3 Charts Analysis

### Chart 1: Total Tax by Month (Area Chart)

[ CHART 9.1 — INSERT HERE ]

**Total Tax by Month**

*Type: Area Chart*

Insert: Power BI screenshot of area chart OR Python matplotlib area chart

| Month | Total Tax (2026) | Trend | Business Interpretation |
|---|---|---|---|
| January | Rs.3,300 | Starting point | Post-holiday recovery; baseline for year |
| February | Rs.3,500 | Peak | Highest tax month — likely salary credit + EMI cycles |
| March | Rs.3,500 (peak) | Peak sustained | Financial year end — maximum transactions |
| April | Rs.3,300 | Decline | New FY start — lower activity; reduced EMIs |

*Table 14: Monthly Tax Trend Analysis (Jan-Apr 2026)*

**Business Insight: Monthly Seasonality**

► Peak tax collection: February-March (Rs.3,500 each) — coincides with financial year-end rush

► March peak driven by: advance tax payments, FY-end EMI clearances, investment transactions

► April decline: new FY start, reduced transaction intensity — predictable seasonal pattern

► Recommendation: Increase customer service capacity in Feb-Mar for peak transaction load

## Chart 2: Total Tax by Transaction Status (Donut Chart)

**[ CHART 9.2 — INSERT HERE ]**

**Total Tax by Transaction Status**

*Type: Donut Chart*

Insert: Power BI screenshot of donut chart OR Python pie/donut chart

| Status | Tax Amount | % Share | Transaction Count | Rate |
|---|---|---|---|---|
| Success | Rs.11.6K | 85.2% | 42,930 | 85.7% of all transactions |
| Failed | Rs.1.5K | 10.9% | 5,103 | 10.2% of all transactions |
| Pending | Rs.0.5K | 3.9% | 2,036 | 4.1% of all transactions |

*Table 15: Transaction Status Distribution — Tax and Count*

**Business Insight: Transaction Failure Rate**

► 85.2% Success rate is below the industry benchmark of 95%+ for mature fintech platforms

► 10.9% Failed transactions (Rs.1.5K tax exposure) = ~500+ failed transactions in 2026 view

- ► 3.9% Pending = Rs.0.5K unrecognised tax — must be settled within regulatory deadlines
- ► Failed transactions by type should be drilled into — Loan EMI failures are highest risk
- ► Recommendation: Investigate top 3 failure reasons; implement retry logic for Failed transfers

## Chart 3: Total Tax by Customer Segment (Horizontal Bar)

**[ CHART 9.3 — INSERT HERE ]**

**Total Tax by Customer Segment**

*Type: Horizontal Bar Chart*

Insert: Power BI screenshot OR Python horizontal bar chart

| Segment | Tax (Rs.) | Customer Count | Tax/Customer | Revenue Priority |
|---|---|---|---|---|
| Retail | 7,000 | 2,703 | Rs.2.59/customer | Highest volume — key mass market |
| Premium | 2,800 | 895 | Rs.3.13/customer | Highest per-customer tax — most valuable |
| SME | 2,200 | 780 | Rs.2.82/customer | Growing segment — target for upgrade |
| Corporate | 1,000 | 374 | Rs.2.67/customer | Fewer customers, moderate per-customer |
| Wealth | 700 | 248 | Rs.2.82/customer | Smallest but high-value — needs retention |

*Table 16: Customer Segment Tax Contribution Analysis*

**Business Insight: Customer Segment Strategy**

- ► Retail (Rs.7K) dominates due to volume (2,703 customers) — mass market drives bulk of tax
- ► Premium segment: highest tax/customer ratio (Rs.3.13) despite lower count — most profitable per head
- ► Wealth segment (248 customers, Rs.700 tax) — smallest but strategic; loss would be significant
- ► SME growing: 780 customers with Rs.2.82/customer — upsell from SME to Corporate is opportunity
- ► Recommendation: Launch Premium upgrade campaigns for top 20% Retail customers; retain all Wealth customers

## Chart 4: Total Tax by State (Horizontal Bar)

[ CHART 9.4 — INSERT HERE ]

**Total Tax by State (Top 10)**

*Type: Horizontal Bar Chart*

Insert: Power BI screenshot OR Python horizontal bar chart

| Rank | State | Tax (Rs.) | Relative Position | Growth Opportunity |
|---|---|---|---|---|
| 1 | Maharashtra | 2,100 | Dominant — 1.5x next state | High penetration; defend market share |
| 2 | Gujarat | 1,700 | Strong second | Growing business hub; increase SME focus |
| 3 | Uttar Pradesh | 1,700 | Tied with Gujarat | Large population; under-served per capita |
| 4 | Karnataka | 1,300 | Tech hub | High-income customers; Premium segment opportunity |
| 5 | Tamil Nadu | 1,300 | Tied with Karnataka | Strong industrial base; Corporate segment |
| 6 | Madhya Pradesh | 1,100 | Mid-tier | Growing; expand branch network |
| 7 | Haryana | 700 | Lower tier | NCR proximity; target Salary class |
| 8 | Kerala | 700 | Lower tier | High literacy; digital channel opportunity |
| 9 | Rajasthan | 700 | Lower tier | Tourism economy; seasonal patterns |
| 10 | Delhi | 600 | Lowest of top 10 | Metro market — highly competitive |

*Table 17: State-wise Tax Analysis with Business Commentary*

**Business Insight: Geographic Strategy**

► Maharashtra (Rs.2.1K) leads by 24% over Gujarat/UP (Rs.1.7K) — Mumbai financial hub effect

► Karnataka + Tamil Nadu tied at Rs.1.3K — IT corridor states with Premium customer potential

► Delhi (Rs.600) surprisingly low despite being the capital — high competition from established banks

► UP (Rs.1.7K) — massive untapped potential; per-capita penetration low despite high tax volume

► Recommendation: Digital-first expansion in Delhi; rural banking products for UP and MP

## Chart 5: Transaction Type Analysis (Matrix Table)

**[ CHART 9.5 — INSERT HERE ]**

**Transaction Type Analysis Matrix**

*Type: Power BI Matrix Table*

Insert: Power BI screenshot of the matrix table from Overview page

| Transaction Type | Count (Total) | Tax (K) | Fee (K) | Amount (M) | Profitability |
|---|---|---|---|---|---|
| Transfer | 8,479 | 0.87K | 3.83K | 12.28M | High fee, moderate tax — profitable |
| Loan EMI | 9,140 | 0.90K | 3.76K | 12.31M | Highest volume + amount — core product |
| Deposit | 4,970 | 0.50K | 1.96K | 6.80M | Moderate — savings building |
| Investment | 3,568 | 0.38K | 1.51K | 5.23M | Low volume, good per-txn value |
| Withdrawal | 4,279 | 0.44K | 0.74K | 2.22M | Low fee — necessary but low margin |
| Card Payment | 5,329 | 0.52K | 0.72K | 2.77M | High count, low fee — everyday txn |
| Bill Payment | 4,310 | 0.44K | 0.58K | 2.15M | Regular — utility of platform |
| Interest Credit | 4,017 | 0.38K | 0.27K | 0.90M | Income for customers; cost for bank |
| Refund | 2,337 | 0.22K | 0.15K | 0.50M | Cost centre — reduces revenue |
| Fee Charge | 3,640 | 0.34K | 0.05K | 0.17M | Pure fee collection — high margin |

*Table 18: Transaction Type Analysis — Count, Tax, Fee, and Amount*

**Business Insight: Transaction Type Profitability**

▶ Loan EMI: highest count (9,140) AND highest amount (Rs.12.31M) — single most important product

▶ Transfer: Rs.3.83K fee on Rs.12.28M — highest fee-earning transaction type; highly profitable

▶ Fee Charge: Rs.3.64K transactions generating Rs.0.05K fee but Rs.0.34K tax — investigate fee structure

▶ Refund: Rs.2.34K refund transactions = revenue reversal; reduce by improving product quality

▶ Interest Credit: Rs.0.90M in credits to customers = bank cost; manage asset-liability ratio

▶ Recommendation: Grow Transfer and Loan EMI volumes; reduce Refund transactions via QA

## Chart 6: Total Tax by Gender (Donut Chart)

**[ CHART 9.6 — INSERT HERE ]**

**Total Tax by Gender**

*Type: Donut Chart*

Insert: Power BI screenshot of gender donut chart

| Gender | Tax Amount | % Share | Customer Count (Raw Data) | Avg Tax/Customer |
|---|---|---|---|---|
| Male | Rs.7.2K | 53.3% | 2,456 customers | Rs.2.93/customer |
| Female | Rs.6.3K | 46.7% | 2,544 customers | Rs.2.48/customer |

*Table 19: Gender-based Tax Contribution Analysis*

**Business Insight: Gender-Based Analysis**

▶ Male customers contribute 53.3% of tax despite being fewer in count (2,456 vs 2,544 Female)

▶ Male avg tax per customer (Rs.2.93) is 18% higher than Female (Rs.2.48) — higher transaction values

▶ Female customer base is larger but generates less tax — opportunity to increase Female engagement

▶ Recommend: Launch women-focused financial products (savings accounts, SIPs); increase Female Loan EMI uptake

# CHAPTER 10: DASHBOARD 2 — TRANSACTIONS PAGE

## 10.1 Dashboard Design and Purpose

The Transactions page provides a detailed drill-through view of individual financial transactions. Unlike the aggregate Overview page, this page enables finance teams, compliance officers, and auditors to inspect specific transactions by customer, type, status, segment, and state. The same KPI cards at the top show filtered totals (e.g., March 2026 filter shows Rs.11.93M, 1.32K transactions).

## 10.2 Filtered KPI State (March 2026 Example)

| 11.93M | 1.32K | 9.06K | 19.72K | 3.55K |
|---|---|---|---|---|
| **Total Amount** | **Total Transactions** | **Avg Txn Value** | **Total Fee** | **Total Tax** |
| *+374.42K vs Prev Year* | *+0.02 vs Prev Year* | *+0.81% vs Prev Year* | *-1.45% vs Prev Year* | *-1.46% vs Prev Year* |

*Table 20: Transaction Page KPIs — Filtered to March 2026*

This filtered view demonstrates the power of the interactive dashboard: a single Year/Month selection instantly recalculates all 5 KPIs, showing March 2026 generated Rs.11.93M (26.3% of full-year Rs.45.32M) from 1,320 transactions. The positive YoY comparison (+Rs.374.42K) for March specifically contrasts with the overall negative YoY (-Rs.539.14K), indicating March 2026 was stronger than March 2025.

## 10.3 Transaction Table Details

| Column | Description | Example Value |
|---|---|---|
| transaction_id | Unique transaction identifier | T00013388 |
| transaction_date | Date of transaction | 24 March 2026 |
| Customer_name | Customer full name (cleaned) | Arjun Jain |
| transaction_type | Type of financial transaction | Fee Charge |
| transaction_status | Outcome: Success/Failed/Pending | Success |
| gender | Customer gender | Female |
| customer_segment | Customer category | SME |

| | | |
|---|---|---|
| state | Customer's home state | Gujarat |
| Total | Total = amount + fee + tax | Rs.0.03 |
| Total tax | Tax component only | Rs.0.01 |
| Total amount | Base transaction amount | Rs.30.27 |

*Table 21: Transaction Table Column Descriptions*

## 10.4 Use Cases for Transactions Page

- Compliance Review: Filter by transaction_status = Failed to identify all failed transactions by customer and type for investigation.
- Customer Query Resolution: Search by Customer_name to see complete transaction history for a specific account.
- Fraud Investigation: Filter high-value transactions with Pending status for manual review and fraud check.
- Segment Analysis: Filter by customer_segment = Wealth to review all high-net-worth customer transactions.
- State Audit: Filter by state = Maharashtra to generate state-specific transaction compliance report.

# CHAPTER 11: PROPOSED SECURITY MECHANISMS

| Level | Mechanism | Implementation |
|---|---|---|
| Data | PII Protection | Customer names anonymised in shared reports; only Customer_ID in external views |
| Data | Fraud Flag Handling | is_fraud column accessible only to Risk team via Row-Level Security |
| Data | Risk Score Masking | risk_score column hidden from standard users; visible only to fraud analysts |
| Access | Row-Level Security | Power BI RLS: state managers see only their state; segment managers see segment |
| Access | Role-Based Access | Viewer: read-only; Analyst: can filter; Admin: full model access |
| Code | Input Validation | Power Query transformations validated before publish; no direct SQL injection path |
| Audit | Change Tracking | Git version control on all PBIX and source file changes |
| Compliance | Data Retention | Transaction data retained per RBI guidelines (7-year minimum) |

*Table 22: Security Mechanisms at Various Levels*

# CHAPTER 12: FUTURE SCOPE AND ENHANCEMENTS

| Enhancement | Description | Technology | Priority |
|---|---|---|---|
| Real-Time Transaction Monitoring | Live dashboard connected to transaction processing system via API | Azure Event Hub, Power BI Streaming | High |
| Fraud Detection ML Model | ML model to predict fraud probability from risk_score + transaction pattern | Python XGBoost, scikit-learn | High |
| Customer Churn Predictor | Identify customers likely to close accounts using transaction frequency drop | Python, Power BI integration | High |
| Credit Scoring Dashboard | Risk-based credit scoring from transaction history and repayment patterns | Python ML, Power BI | Medium |
| Multi-Currency Support | Handle international transactions with forex conversion in DAX | Power BI DAX, API | Medium |
| Mobile App Dashboard | Power BI Mobile app for executives to monitor KPIs on the go | Power BI Mobile, Azure | Medium |
| NLP Complaint Classifier | Auto-classify customer complaints and failed transaction reasons using NLP | Python BERT/NLTK | Low |
| WhatsApp Alert Bot | Automated WhatsApp alerts for large transactions, failures, or fraud | Twilio API, Azure Functions | Low |

*Table 23: Future Enhancement Roadmap*

# CHAPTER 13: CONCLUSION

This project has successfully delivered FinSight — a two-page interactive Finance Analytics Dashboard that transforms 55,069 raw financial records into a production-grade business intelligence tool. Starting from two CSV files with data quality issues (mixed name casing, duplicate channel labels, string-format dates), the project executed a complete analytics pipeline including Power Query cleaning, DAX measure development, and a dual-page Power BI dashboard delivering 12 visualisations and 4 interactive slicers.

The Overview Analysis page reveals five critical business insights. First, Total Amount declined by Rs.539.14K YoY, signalling revenue contraction that requires immediate management attention. Second, Loan EMI dominates as both the highest-volume (9,140 transactions) and highest-amount (Rs.12.31M) transaction type, confirming it as the core product requiring investment in quality and capacity. Third, Maharashtra leads state revenue at Rs.2.1K tax with Gujarat and UP tied at Rs.1.7K, but Delhi's low contribution (Rs.600) despite being the capital reveals a competitive penetration challenge. Fourth, the 10.9% Failed transaction rate (well above the 5% industry benchmark) represents significant revenue leakage and customer dissatisfaction requiring urgent investigation. Fifth, the Premium customer segment, despite ranking second in total tax (Rs.2.8K), delivers the highest tax-per-customer ratio (Rs.3.13), making them the most profitable group for targeted retention and upsell.

The Transactions page provides the compliance and customer service teams with an auditable, filterable record of all individual transactions. When filtered to March 2026, it shows Rs.11.93M from 1,320 transactions — 26.3% of the annual total — confirming March as the peak month, consistent with financial year-end patterns observed in the monthly tax area chart.

From an academic perspective, this project demonstrates mastery of the complete Business Intelligence lifecycle: requirements gathering from a formal Business Requirements document, data cleaning via Power Query, relational data modelling, advanced DAX measure development including YoY time intelligence, interactive dashboard design with dynamic metric switching via Field Parameters, and professional documentation. The FinSight dashboard is a portfolio-quality deliverable demonstrating skills directly applicable to Financial Analyst, Business Intelligence Developer, and Data Analyst roles in banking, fintech, and financial services.

# CHAPTER 14: BIBLIOGRAPHY


## Books

[1] Colgan, A. (2021). Supercharge Power BI: Power BI is Better When You Learn to Write DAX. Holy Macro! Books.

[2] Kimball, R., & Ross, M. (2013). The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling (3rd ed.). John Wiley & Sons.

[3] McKinney, W. (2022). Python for Data Analysis (3rd ed.). O'Reilly Media.

[4] Ferrari, A., & Russo, M. (2019). The Definitive Guide to DAX (2nd ed.). Microsoft Press.

## Industry Reports

[5] Deloitte. (2023). Tech Trends 2023: Analytics and AI-Driven Enterprises. Deloitte Insights.

[6] McKinsey Global Institute. (2020). The Data-Driven Enterprise of 2025. McKinsey & Company.

[7] Reserve Bank of India. (2023). Report on Trend and Progress of Banking in India 2022-23. RBI Publications.

[8] Gartner. (2021). Magic Quadrant for Analytics and Business Intelligence Platforms. Gartner Research.

## Online Resources

[9] Microsoft Corporation. (2024). Power BI Documentation. https://docs.microsoft.com/power-bi/

[10] DAX Guide. (2024). DAX Functions Complete Reference. https://dax.guide/

[11] Microsoft Learn. (2024). Power Query M Formula Language. https://learn.microsoft.com/power-query/

[12] Pandas Documentation. (2024). https://pandas.pydata.org/docs/

[13] Towards Data Science. (2024). Finance Analytics and Power BI Articles. https://towardsdatascience.com/

[14] SQLBI. (2024). DAX Patterns for Finance. https://www.daxpatterns.com/